

NMR measure of Translational Diffusion and Fractal Dimension. Application to molecular mass measurement.

- Supporting Information -

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This manuscript is dedicated to Sophie Augé, deceased june 21st 2006

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A. Experimental Details

A.1. Protein study

Protein data were collected on a Bruker AMX-600 spectrometer with a field gradient accessory capable to deliver z-field gradients up to 50 Gcm⁻¹. In all NMR experiments, the probe temperature was maintained at 22.8°C and 5 mm sample tubes were used. The LED [6] spin-echo pulse sequence with bipolar field gradients [15] was used for all the data reported. Measurement of ¹H NMR spectra in aqueous solution required the suppression of the large resonance from H₂O and we used a modified version of the LED sequence with a WATERGATE scheme included after the final reading pulse [11].

Sequence delays were $\Delta = 200$ ms (diffusion delay), $\tau = 1.2$ ms (after gradient recovery delay), and $T_e = 3$ ms (LED recovery delay). For each data set, 2048 complex points were collected for each 35 experiments in which the gradient strength was exponentially incremented from 1 to 47.5 Gcm^{-1} [1]. The gradient duration $\delta/2$ was adjusted to observe a near complete signal loss at 47.5 Gcm^{-1} . Typically, the $\delta/2$ delay was chosen in the 0.7 - 1.5 ms range. A 1 s recycle delay was used between scans for all data shown. In all cases, the WATERGATE gradients were kept at 4 Gcm^{-1} for 1 msec, and the total acquisition time ranged from 30 minutes to 14 hours depending on the sample.

We purchased BPTI (Bovine Pancreatic Trypsin Inhibitor, Sigma), ubiquitin (Sigma), cytochrome C (Sigma), ribonuclease A (Pharmacia), lysozyme (Boehringer Mannheim), myoglobin (Sigma), β -lactoglobulin A (Sigma), chymotrypsinogen A (Pharmacia), and ovalbumin (Sigma) as lyophilized salt-free powders. The Glutathione S-Transferase (GST) was released from GST-fusion protein by thrombin cleavage. Charybdotoxin [2], cardiotoxin- γ [7], and neurotoxin α [16] were obtained from A. Ménez. Thioredoxin was obtained from F. de Lamotte-Guéry as presented in [4]. Porine MOMP was obtained from G. Labesse as presented in [9].

All the proteins having a molecular weight higher than 8 kD were desalted with the following protocol. A quantity of 2 mg of protein was dissolved in 3 ml of 50 mM phosphate buffer (pH 5.8) and centrifuged at 10°C , 4000 g in a 4 ml tube (membrane Biomax-5, Ultrafree-4, Millipore). The sample was concentrated again to approximately 200 - 400 μl . Three wash cycles were applied. Protein concentrations were determined by UV absorbance measurement at 280 nm wavelength (410 nm for myoglobin). All samples were prepared for a protein concentration of 3 mg/ml in a phosphate buffer (50 mM, pH 5.8, in water ($\text{H}_2\text{O}:\text{D}_2\text{O}$, 90%:10%) containing 0.5 mM of Tris.

For each data set, the spectral axis was processed with sine-bell, and Fourier transform was applied in order to obtain 1024 real points. A spline baseline correction was finally applied. The columns (axis of varying gradient) of this data-set were then processed for inverse Laplace transform using the Inverse Laplace technique [5] using the *Gifa* software [12]. The DOSY reconstruction was realized with 490 points in the diffusion dimension and 2000 MaxEnt iterations. For each protein, the diffusion coefficient was calculated as the mean value of the diffusion coefficients observed for the 15 most intense columns in the 0.22 - 2.46 ppm region. The uncertainty on the measure was estimated from the standard deviation σ on these values, and chosen at $\pm\sigma$ (95% confidence level). For each data set, the diffusion coefficient of the Tris molecule was calculated from the column at 3.70 ppm. The results are summarized in 1.

A.2. Polymer study

Polymers were measured either on a 600 MHz (PS in toluene) or on a 400 MHz spectrometer (other studies). Polyethyleneoxide (PEO) and polystyrene (PS) calibrated monodisperse polymers were purchased from Polymer Labs or Amerian Polymer Standards Corporation, the reported polydispersity index being typically between 1.02 and 1.10. Materials were used as received. PEO standards were stored in a refrigerator to minimize degradation. Measures were performed at 30°C . The diffusion time was typi-

protein	mol. weight (kD)	Prot. Diff. ($\mu\text{m}^2\text{s}^{-1}$)	Tris Diff. ($\mu\text{m}^2\text{s}^{-1}$)
Charybdotoxin	4.190	148.8 ± 2.2	530.0
BPTI	6.517	120.1 ± 5.2	491.0
Neurotoxine α	6.795	116.3 ± 3.1	533.0
Cardiotoxine γ	6.827	116.1 ± 3.9	560.0
Ubiquitine	8.565	111.4 ± 1.9	513.0
Thioredoxine	11.600	101.8 ± 8.6	530.0
Cytochrome C	12.380	99.8 ± 1.2	520.0
Ribonuclease A	13.700	94.1 ± 1.6	553.0
Lysozyme	14.600	95.0 ± 2.0	519.0
Myoglobine	16.900	89.3 ± 2.9	561.0
β -lactoglobuline A	17.675	81.5 ± 5.2	558.0
Chymotrypsinogen A	25.000	76.0 ± 2.0	570.0
Ovalbumine	43.000	60.1 ± 1.8	569.0
Glutathione			
S transferase <i>dimer</i>	52.000	55.7 ± 3.6	548.0
MOMP	120.000	47.0 ± 2.5	570.0

Table 1: list of the protein measurements performed in this study

cally 50 ms, increased to a range of 100 to 350 ms for the analysis of ultra-high molecular mass polymers, such as the 7,800 and 18,000 kDa PS samples. The number of scan was adapted to the sample, but never less than eight scans. The total experiment time, including a 4 min temperature equilibration step, ranged from 15 minutes to several hours per sample

All experiment (except for PS in toluene) were acquired as follows. The ^1H PFG NMR diffusion experiments utilized a Bruker Avance-400 spectrometer operating at a ^1H frequency of 400.1 MHz, equipped with a 5 mm multinuclear broadband-observe z-axis gradient probe capable of generating 55 G/cm field strengths. The PFG NMR experiments employed a convection-compensating double stimulated-echo pulse sequence [8] with three 0.6-ms sinusoidal-shaped spoiler gradient pulses. Smoothed-square gradient pulses with durations typically in the range of 4-14 ms were used for each of the four diffusion pulses. eight spectra were obtained using varying gradient strengths.

Spectra were processed with 3 Hz Lorentzian line-broadening using the TopSpin software (Bruker, Karlsruhe, Germany) and an automatic baseline correction routine was then applied. Manual baseline refinement had to be conducted in some cases.

PS in toluene and linear alkanes were acquired as follows. The ^1H PFG NMR diffusion experiments utilized a Unity-Varian 600 spectrometer operating at a ^1H frequency of 600.1 MHz, equipped with a 5 mm $^1\text{H} - \text{X}$ reverse z-axis gradient probe capable of generating 60 G/cm field strengths. The PFG NMR experiments employed the Oneshot [10] experiment a with κ coefficient of 0.2. Square gradient pulses with durations typically in the range of 1.5-4 ms were used for each of the four diffusion pulses.

Spectra were processed with 3 Hz Lorentzian line-broadening using the NMRNoteBook

software (NMRtec, Illkirch France), and an automatic baseline correction routine was then applied. Manual baseline refinement had to be conducted in some cases.

Diffusion coefficients were estimated from the decay of the polymer signal either using direct fit of the gaussian attenuation law, or by Inverse Laplace [5] using the *Gifa* software [12].

molecular family	reference	C_r	d_F
Linear alkanes	toluene	26.5 ± 5.6	1.41 ± 0.08
PEO in D ₂ O	HDO	5.00 ± 0.13	1.856 ± 0.010
PEO in CDCl ₃	TMS	12.36 ± 0.31	1.7342 ± 0.0085
PS in acetone	TMS	6.4 ± 1.5	2.15 ± 0.13
PS in CDCl ₃ below 20 kD	TMS	6.93 ± 0.54	2.117 ± 0.042
PS in CDCl ₃ above 20 kD	TMS	28.0 ± 3.2	1.630 ± 0.024
PS in toluene	toluene	4.30 ± 0.44	2.446 ± 0.070
PS in THF below 30 kD	TMS	8.09 ± 0.80	2.014 ± 0.048
PS in THF above 30 kD	TMS	29.3 ± 2.7	1.608 ± 0.018
PMMA in CDCl ₃ below 30 kD	TMS	8.04 ± 0.72	2.069 ± 0.042
PMMA in CDCl ₃ above 25 kD	TMS	28.1 ± 2.7	1.649 ± 0.022
PMMA in acetone below 30 kD	TMS	7.5 ± 1.2	2.174 ± 0.080
PMMA in acetone above 25 kD	TMS	17.3 ± 1.8	1.836 ± 0.030
small molecules in D ₂ O [3]	HDO	5.17 ± 0.59	1.836 ± 0.072
small molecules in CDCl ₃ [3]	TMS	11.92 ± 0.72	1.772 ± 0.035
globular proteins	Tris	7.2 ± 1.3	2.56 ± 0.13
oligosaccharides [13]	HDO	3.77 ± 0.34	2.071 ± 0.039
denatured peptide [14]	dioxan	16.1 ± 4.4	1.715 ± 0.091

Table 2: measured values for C_r and d_F (as defined in equation 6 in text) for all conditions and molecular families.

B. Molecular mass determination

B.1. Data analysis

The data were analyzed with a home written scipy/python program freely available upon from the authors request. In the cases of oligosaccharides and denatured peptides, the published values were used. d_F and C_r values were obtained from a direct weighted fit of equation 6 on the data, which did not require to take the log of the analysed values. Complete error analysis was performed as detailed in the next section. When no experimental errors were available, it was assumed that the diffusion coefficients were measured with a 10% error bar. Results are shown in 2.

Obtained R coefficients ranged from 0.995 to 0.99999 for polymer studies, R was 0.976 for proteins, 0.977 for small molecules in water, 0.983 for small molecules in CDCl₃, 0.992 for linear alkanes, 0.994 for denatured peptides, and 0.9994 for oligosaccharides. Uncertainties are given by a covariance matrix analysis.

C. Uncertainty Calculus

$u(a)$ denotes the uncertainty of the quantity a and $\text{cov}(a, b)$ denotes the covariance between quantities a and b .

C.1. uncertainty on D_r

$$D_r = \frac{D}{D_{\text{solv}}} \quad (1)$$

then

$$u(D_r) = \frac{u(D)}{D_{\text{solv}}} + u(D_{\text{solv}}) \frac{D}{D_{\text{solv}}^2} \quad (2)$$

C.2. uncertainty on M

if

$$M = \left(\frac{C_r}{D_r} \right)^{d_F} \quad (3)$$

given the law of propagation of uncertainty :

for $y = f(x_i)$

$$u^2(y) = \sum_i^N \left(\frac{\partial f}{\partial x_i} \right)^2 u^2(x_i) + 2 \sum_i^{N-1} \sum_{j=i+1}^N \frac{\partial f}{\partial x_i} \frac{\partial f}{\partial x_j} \text{cov}(x_i, x_j)$$

then

$$u^2(M) = \left(d_F \frac{M}{D_r} \right)^2 u^2(D_r) \quad (4)$$

$$+ \left(d_F \frac{M}{C_r} \right)^2 u^2(C_r) \quad (5)$$

$$+ M^2 \ln^2 \left(\frac{C_r}{D_r} \right) u^2(d_F) \quad (6)$$

$$+ 2 d_F \frac{M^2}{C_r} \ln \left(\frac{C_r}{D_r} \right) \text{cov}(C_r, d_F) \quad (7)$$

These expressions are used in the program determining M from D , D_{solv} and parameters from 2.

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